

“So, you want to do experiments on the Cloud?”

Rencontres du PEPR Cloud 2026 – Nantes

Quentin Guilloteau, Inria, Lyon

30-09-2026

What is this talk about? $\rightsquigarrow$ “Reproducibility” and “Doing Experiments”

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What even is “Reproducibility”?

- It depends on who you ask...
- In computer science: ACM gave some definitions (changed in 2020)
 - *Repeatability* (Same team, same experimental setup)
 - *Reproducibility* (Different team, same experimental setup)
 - *Replicability* (Different team, different experimental setup)
- some sciences use also the term “Robustness”

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What I think we want/need

Precise introduction of Variation $\rightsquigarrow$ extend work of others

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“Reproducibility”:
Result of Scientific/Experimental Methodology

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What I think we want/need

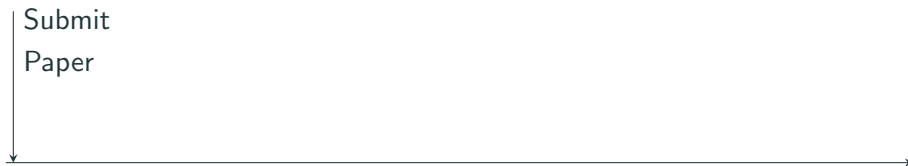
Precise introduction of Variation $\rightsquigarrow$ extend work of others

Artifact Evaluation (AE)



The Artifact Evaluation Process in a Nutshell

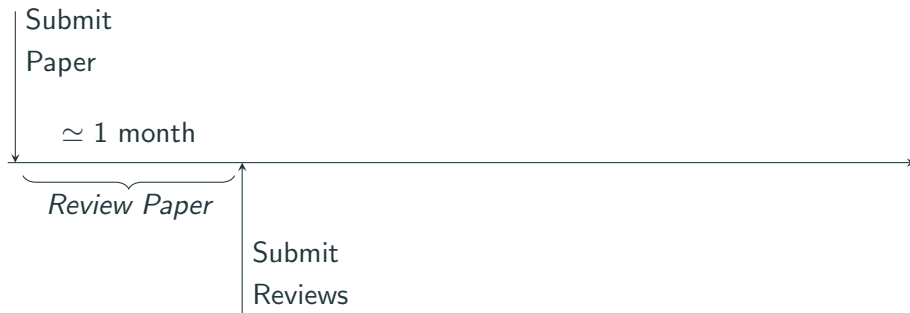
Authors



Timeline not to scale!

The Artifact Evaluation Process in a Nutshell

Authors

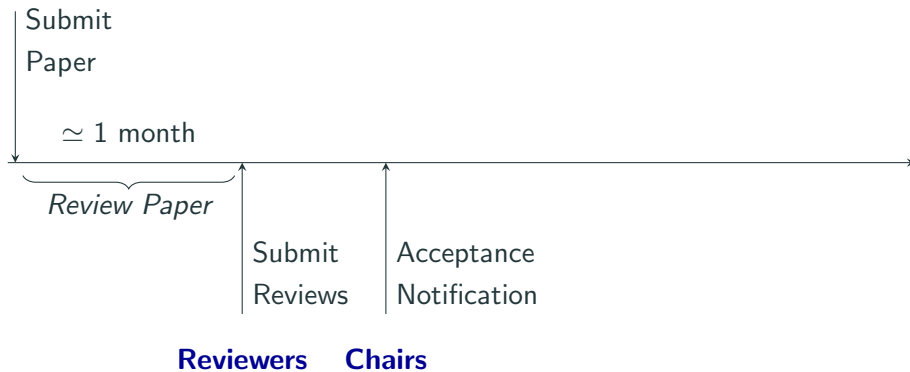


Reviewers

Timeline not to scale!

The Artifact Evaluation Process in a Nutshell

Authors



Timeline not to scale!

The Artifact Evaluation Process in a Nutshell

Authors

Submit
Paper

$\simeq 1$ month

Review Paper

Reviewers

Authors

Submit
Artifact

Submit
Reviews

Acceptance
Notification

Chairs

Timeline not to scale!

Appendix: Artifact Description/Artifact Evaluation

Artifact Description (AD)

I. OVERVIEW OF CONTRIBUTIONS AND ARTIFACTS

A. Paper's Main Contributions

Provide a list of all main contributions of the paper.
 C_1 This is the 1st contribution.
 C_2 This is the 2nd contribution.
 C_3 This is the 3rd contribution.

B. Computational Artifacts

List the computational artifacts related to this paper along with their respective DOIs. Note that all computational artifacts may be archived under a single DOI.

A_1 <https://doi.org/YYYYY/nomads.0XXXXX>

A_2 <https://doi.org/ZZ.YYYY/nomads.1XXXXX>

A_3 <https://doi.org/ZZ.YYYY/nomads.2XXXXX>

Provide a table with the relevant computational artifacts, highlight their relation to the contributions (from above) and point to the elements in the paper that are reproducible by each artifact, e.g., which figures or tables were generated with the artifact.

Artifact ID	Contributions Supported	Related Paper Elements
A_1	C_1	Tables 1-2 Figure 3
A_2	C_2	Tables 2-3 Figures 1-2
-	-	-

II. ARTIFACT IDENTIFICATION

Provide the following six subsections for each computational artifact A_i .

A. Computational Artifact A_i

Relation To Contributions

Briefly explain the relationship between the artifact and contributions.

Expected Results

Provide a higher level description of what outcome to expect from the corresponding experiments. Provide an explanation of how the results substantiate the main contributions.

Algorithm A should be faster than Algorithms C and B in all GPU scenarios.

Expected Reproduction Time (in Minutes)

Estimate the time required to reproduce the artifact, providing separate estimates for the individual steps: Artifact Setup, Artifact Execution, and Artifact Analysis.
The expected computational time of this artifact on GPU X is 20 min.

Artifact Setup (incl. Inputs)

Hardware: Specify the hardware requirements and dependencies (e.g., a specific interconnect or GPU type is required).

Software: Introduce all required software packages, including the computational artifact. For each software package, specify the version and provide the URL.

Datasets / Inputs: Describe the datasets required by the artifact. Indicate whether the datasets can be generated, including instructions, or if they are available for download, providing the corresponding URL.

Installation and Deployment: Detail the requirements for compiling, deploying, and executing the experiments, including necessary compilers and their versions.

Artifact Execution

Provide an abstract description of the experiment workflow of the artifact. It is important to identify the main tasks (processes) and how they depend on each other.

A workflow may consist of three tasks: T_1 , T_2 , and T_3 . The task T_1 may generate a specific dataset. This dataset is then used as input by a computational task T_2 , and the output of T_2 is processed by another task T_3 , which produces the final results (e.g., plots, tables, etc.). State the individual tasks T_i and provide their dependencies, e.g., $T_1 \rightarrow T_2 \rightarrow T_3$.

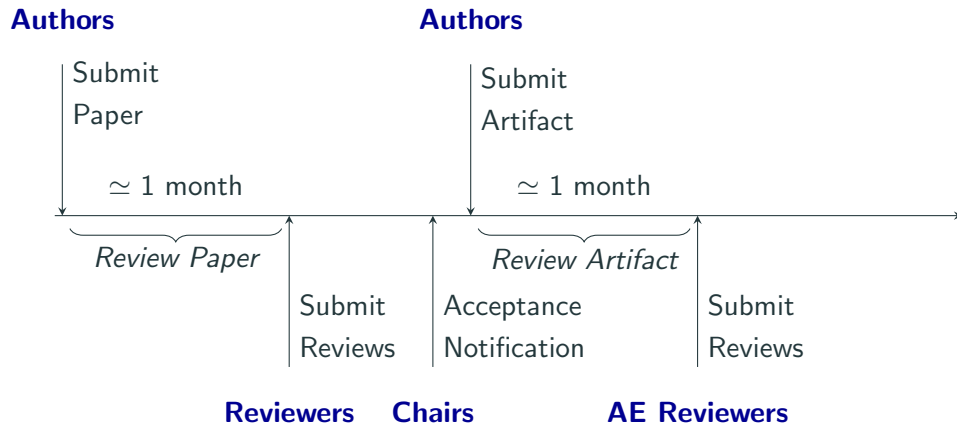
Provide details on the experimental parameters. How and why were parameters set to a specific value (if relevant for the reproduction of an artifact, e.g., size of dataset, number of data points, input sizes, etc.). Additionally, include details on statistical parameters, like the number of repetitions.

Artifact Analysis (incl. Outputs)

B. Computational Artifact A_i

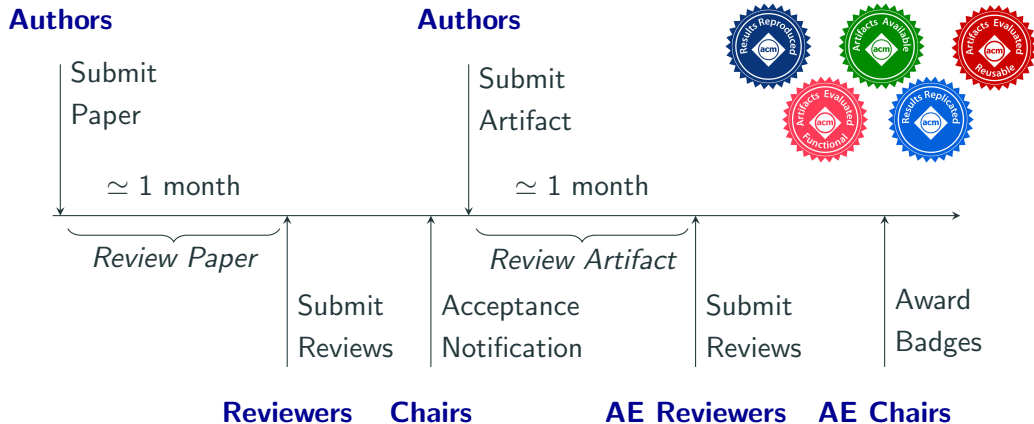
Provide the same type of information as done for Computational Artifact A_1 .

The Artifact Evaluation Process in a Nutshell



Timeline not to scale!

The Artifact Evaluation Process in a Nutshell



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Incoming: My therapy session



Timing, Communication, and Resubmission

Timing of AE

- Artifacts **quickly** created by the authors close to the deadline
- AE Reviewers have a short time to evaluate in a **busy** schedule
- **Available** physical resources might be hard to find during evaluation process

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Communication Between Authors and AE Reviewers

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Resubmission Not Possible

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Communication Between Authors and AE Reviewers

Does the AE process need to be **redesigned**?
Only for journals? Required **before** submission?

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Longevity and AE Reports

Longevity

- Science: **Iterative Self-Correcting Process**
(“standing on the shoulders of giants” – Isaac Newton)

Longevity and AE Reports

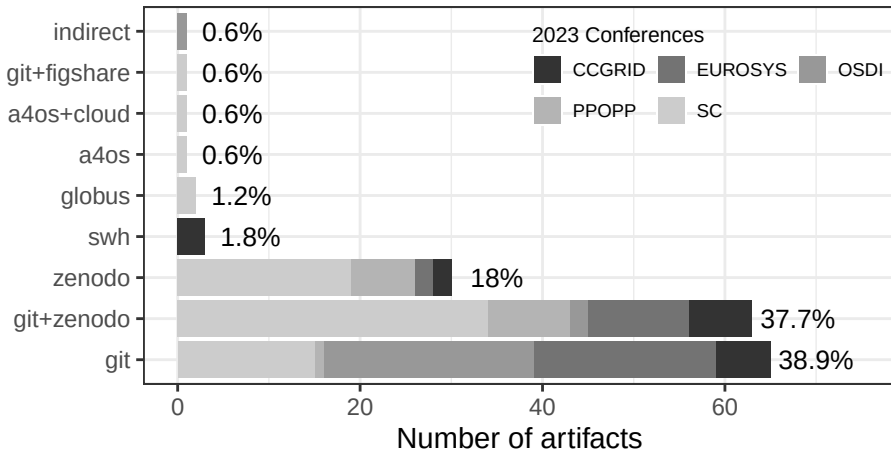
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Longevity and AE Reports

Longevity

- S
- (
- R



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Git, yes, but...

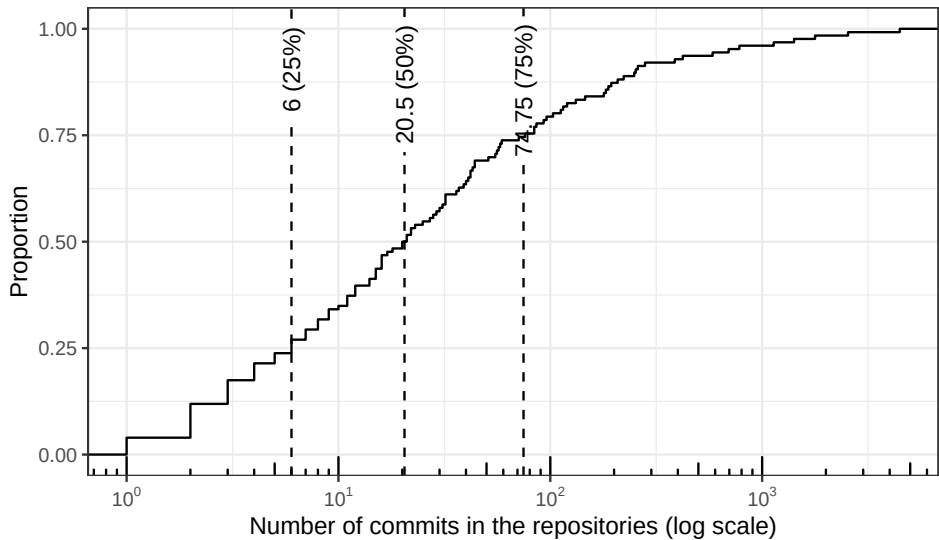
Showing 15,115 changed files with 6,755,080 additions and 1 deletion.

~ Zenodo, Software-Heritage

Longevity and AE Reports

Longevity

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Longevity and AE Reports

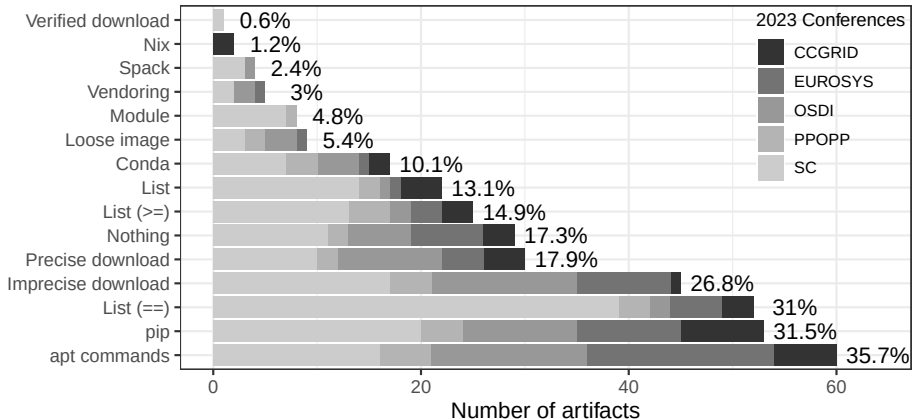
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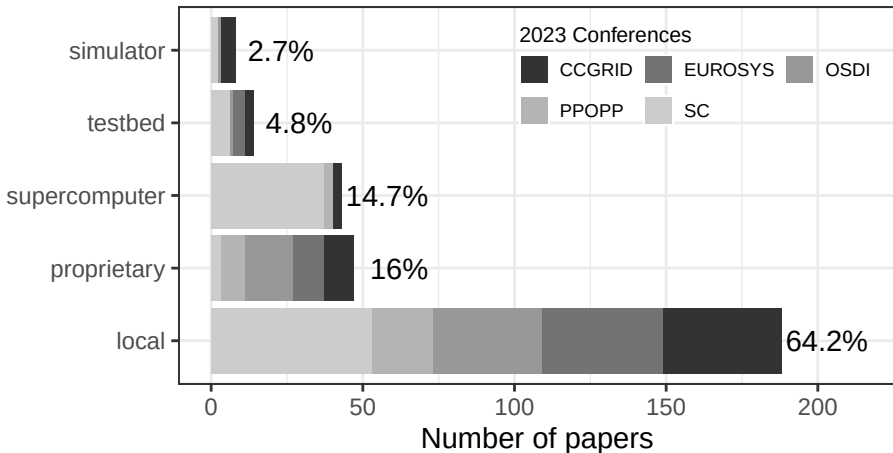
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 - Poor control of software environments
(package list, sometimes versions, apt, pip, userspace only)
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⇒ Experimental testbeds

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↪ **For who** is reproducible research?

AE Reports

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- Badges $\simeq$ pass/fail ($\rightsquigarrow$ a score?)
- Review reports might contain complementary information to AD and badges
 $\rightsquigarrow$ **no access** for future researchers
- AE Reports allow for **more details** about the reviewers' reproduction attempts

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↪ Do badges carry **enough nuance/information?**

Longevity and AE Reports

Longevity

↪ **For who** is reproducible research?

Who should benefit from AE?

Authors and/or future researchers?

- Review reports might contain complementary information to AD and badges
↪ **no access** for future researchers
- AE Reports allow for **more details** about the reviewers' reproduction attempts

↪ Do badges carry **enough nuance/information**?

Software Environments

What is Computation? (from Konrad Hinsén's slides)

Input

```
100111100001001100110101101100
00101001110101011110001001101
01011110110001110111011110001
001100001110111000100100000111
110101100111001110100000100110
11011110011100001111101101111
111001001011110001100110000101
011100001000010001011110000010
110101110011101111001010100111
111000101110011001101101001001
011001010100101011000001001100
110100111001011111100001011101
01111011111000111011110101101
000001110110011001010101011100
100010110001100000111001100010
000000111011100100100101010111
000010000001100001000010110110
101111101111000111100101110101
100101010100001001110100010001
011110011010100101111011110101
100011000110110001011101100110
110100000100000011011000001101
100000011100100111101101011011
010110010001000101110111001010
```



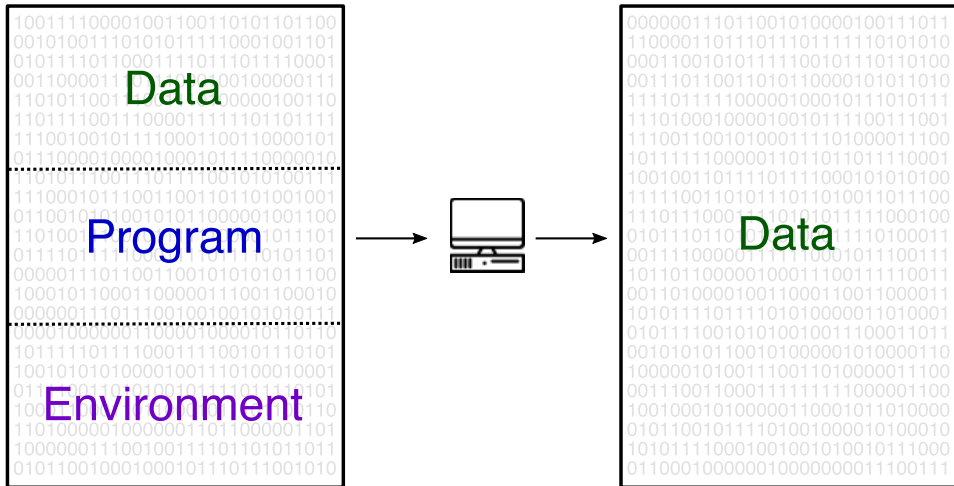
Output

```
000000111011001010000100111011
110000110111011101111110101010
00011001010111110010111010100
001110110011010110000101011010
111101111100000100010111010111
111010001000010010111100111001
111001100101000111101000011100
101111110000011011011011110001
100100110111101111000101010100
111110011010111011010011011100
111011100011110101011111000100
010111011010100100011110100011
001111000001111110001011100111
101101100000100011100111110011
001101000010011000110011000011
101011110111101010000011010001
010111100101010010011100011011
001010101100101000001010000110
100000101001110011010000011100
001110011000111111111000001100
100100010100000110001011010000
010110010111101001000010100010
101011110001001001010010111000
011000100000010000000011100111
```

What is Computation? (from Konrad Hinsén's slides)

Input

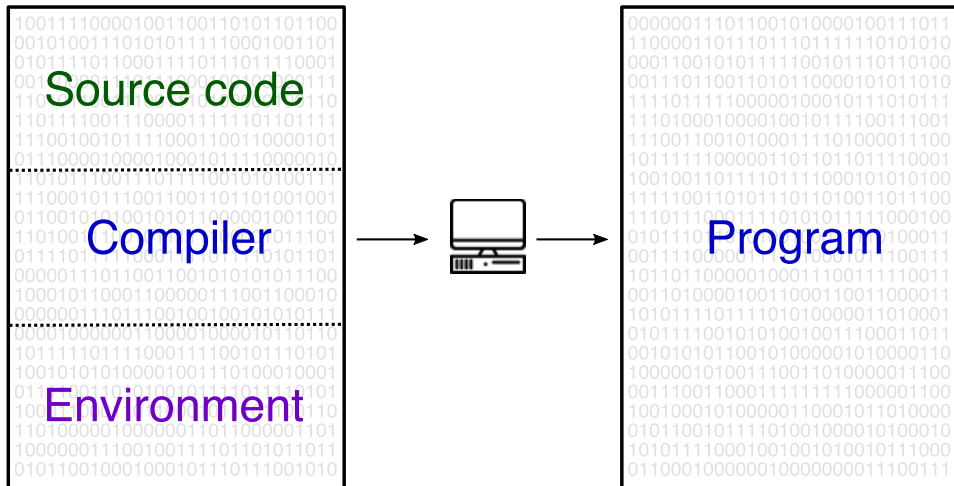
Output



What is Computation? (from Konrad Hinsén's slides)

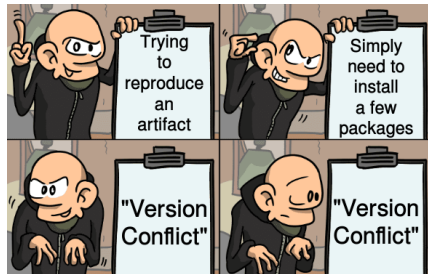
Input

Output



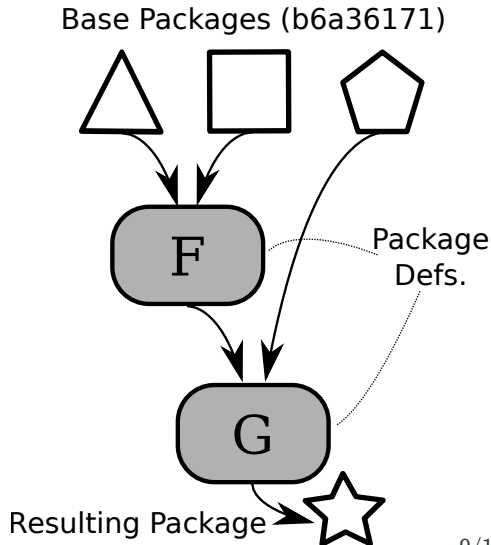
Software Environment – The usual mistakes

- Depend on an **uncontrollable external** state
(`apt update`, `apt install`)
- Check the content of downloaded objects
 - The object behind the same URL **can change through time**
 - Take the cryptographic hash of the object
(`sha256sum`, `md5sum`, etc.)
- Pay attention to the commit for Git repositories
(`git clone https://github.com/me/myrepo`)
 - By default: **latest commit** of main branch
 - Commit can be deleted... (Software Heritage)
- Classical Package Managers put everything in
`/usr/bin`, `/usr/lib` $\rightsquigarrow$ **potential conflicts**



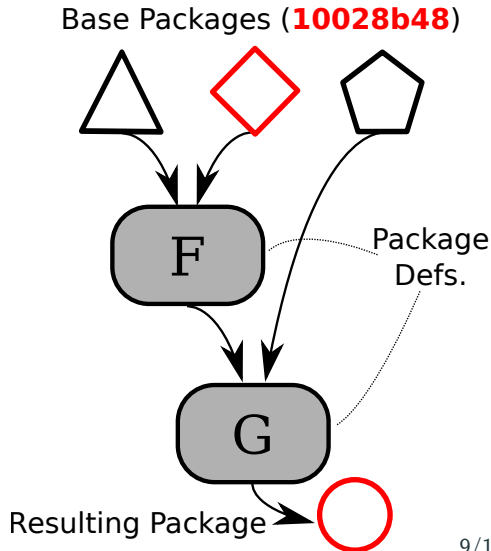
Functional Packages Managers

- Guix, Nix: reproducible by design!
- **packages = functions**
 - inputs = dependencies
 - body = commands to build the package
 - base packages defined in Git
- sandbox, **no side effect**
- `/gnu/store/hash(inputs)-my-pkg`
- Store **immutable, read-only**
- precise definition of `$PATH`
- can build: container, VM, etc.



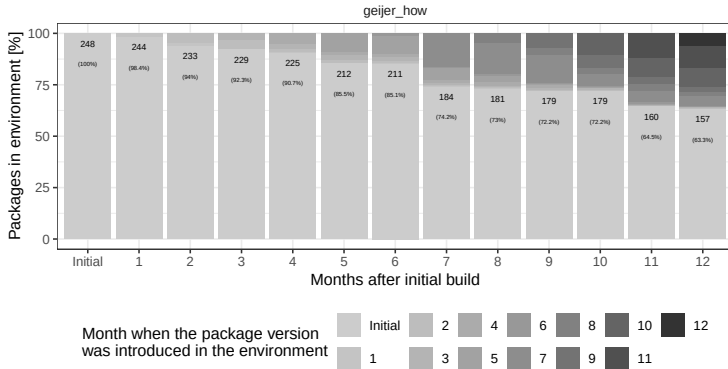
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“Yes, but Docker ...” No!

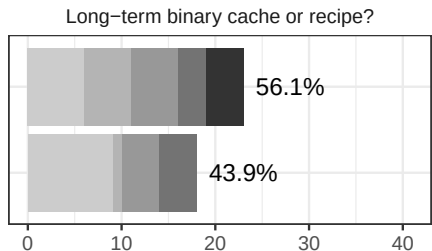
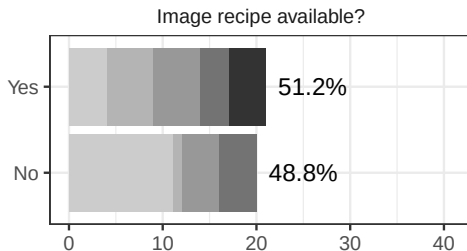
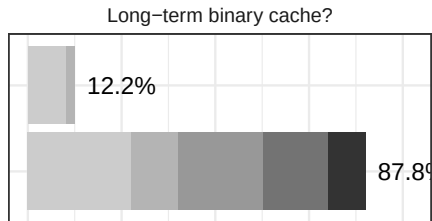
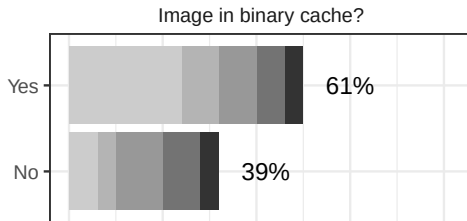
Evolution of the packages versions over time for selected Dockerfile studied



- Took Dockerfiles from artifacts
- Rebuilt them every month for a year

- $\simeq 25\%$ of the environment changed!
- Malka et al.

“Yes, b



Number of artifacts

2023 Conferences



CCGRID



EUROSYS



OSDI



PPOPP



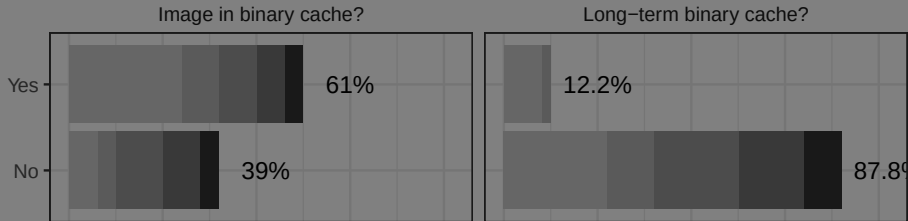
SC

• To

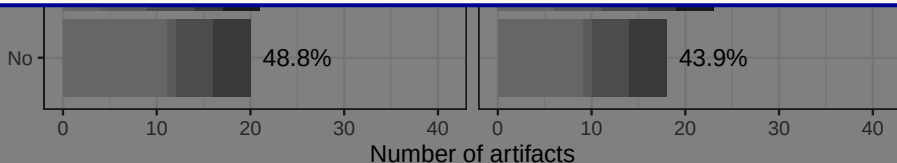
• Re

nged!

“Yes, b



Do we want to store obscure binaries or text files?
Remember: we want precise variation



2023 Conferences



CCGRID



EUROSYS



OSDI



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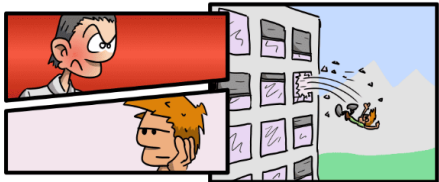
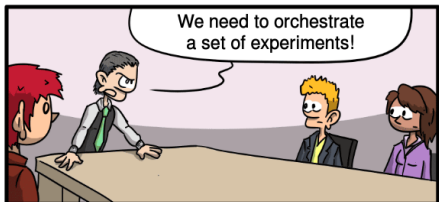


SC

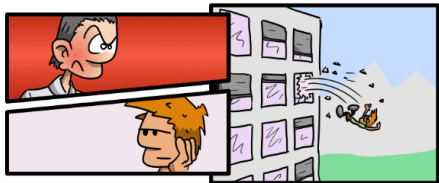
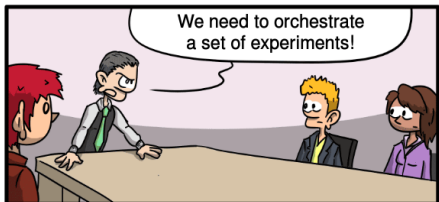
nged!

Workflow Managers

How to define and orchestrate a set of experiments?



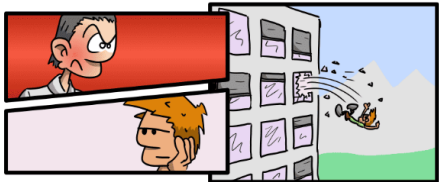
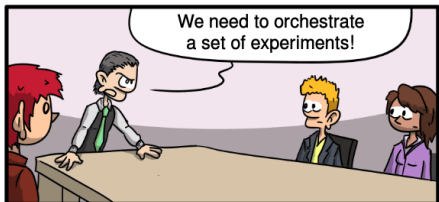
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Workflow Managers

- Explicit definitions of steps + deps
- Efficient concurrent execution
- Partial (re)builds
- Examples: Snakemake, Nextflow, CWL, Make

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Workflow Managers

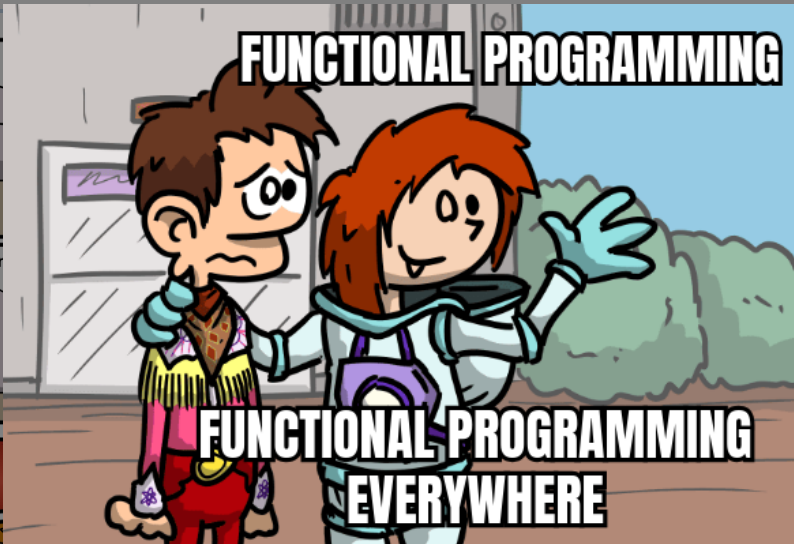
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- Examples: Snakemake, Nextflow, CWL, Make

```
rule run_expe:
    input:
        run="src/run.py"
        dat="data/trace.owf"
    output:
        "results/{foo}/{bar}/{iter}.csv"
    shell: """
        python3 {input.run} -data {input.dat}
        -foo {wildcards.foo}
        -bar {wildcards.bar}
        -out {output}
    """
```

How to define and orchestrate a set of experiments?



Let's just use a mega bash script



eps

y, CWL, Make

.csv"

[input.dat}

```
-out {output}  
"""
```

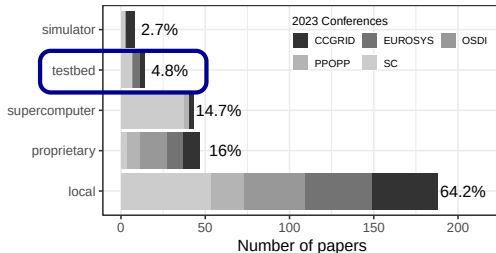
Hardware and Testbeds

Hardware and Experiment Platforms

Testbeds (SLICES-RI, Grid'5000, Chameleon, CloudLab) are **amazing** to access hardware and reduce unknown/variability deployment

But...

- Waiting Time > AE Review Time
- Not widely used by the authors
- Strong dependency on the testbed?
(Kadeploy3 update: thanks Claude...)

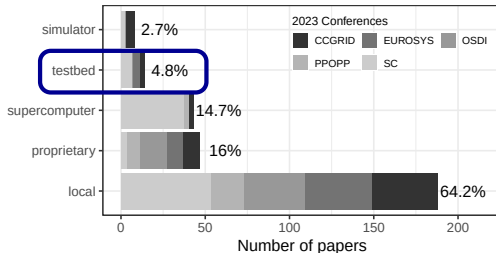


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- **Realism**: Cutting-edge + Scale = \$\$\$



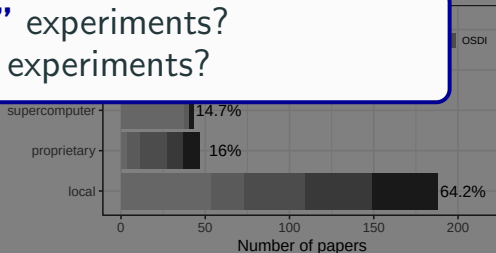
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But

How to **“future proof”** experiments?
How to have **realistic** experiments?

-
- Not widely used by the authors
- Strong dependency on the testbed?
(Kadeploy3 update: thanks Claude...)



Conclusion



Conclusion – Supply and Demand of Research Artifacts

We have “a lot” of artifact **supply**, but not much **demand**...

Others “Demand” Opportunities

- Reproducibility Hackathons
- Teaching



Figure 1: ReproHackathon at COMPAS26

Conclusion – Supply and Demand of Research Artifacts

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Others “Demand” Opportunities

- Reproducibility Hackathons
- Teaching
- **Extent the work of others**

Community: be a Giant

- Database of experiments
- “Don’t reinvent the wheel”
- Kickstart the research of PhD students



Figure 1: ReproHackathon at COMPAS26

Experimental research is hard: but we can do better!

For PhD Students?

- Participate in AE/JOSS
- Take your time and discuss with people

For Researchers?

- Encourage your students
- Organize events (team, classes)

For Testbeds?

- Make expes easily sharable and searchable?
- Stay amazing

Conclusion – Perspectives

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- Stay amazing

For the community

Organize, share, promote, discuss, exchange, help, **slow down**

Conclusion – Perspectives

Experimental research is hard: but we can do better!

For PhD Students?

- Participate in

For Researchers?

- Encourage your

For Testbeds?

- Make expes easily

Reproducibility is **not** a technical problem anymore

For the community

Organize, share, promote, discuss, exchange, help, **slow down**

Advertisement! – GTEB SLICES-FR

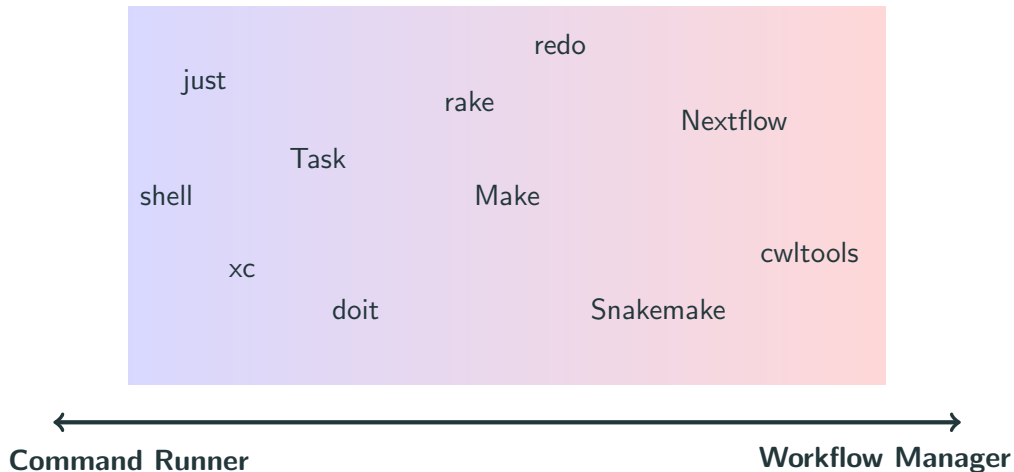
- Groupe Thématique d'Expression de Besoins
- **“Outils et méthodes pour l'expérimentation”**
- Discussing those questions in the context of SLICES-FR
- Do you have experimental needs?
- Feel free to join and discuss! (can also come talk to me)

Mailing List

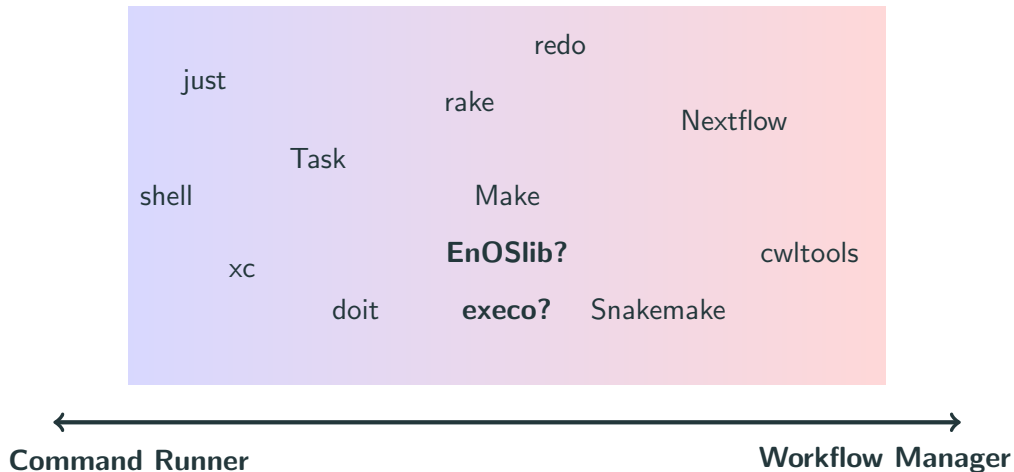
<https://groupes.renater.fr/sympa/info/slices-fr-course-gteb-experimentations>

slices-fr-course-gteb-experimentations@groupes.renater.fr

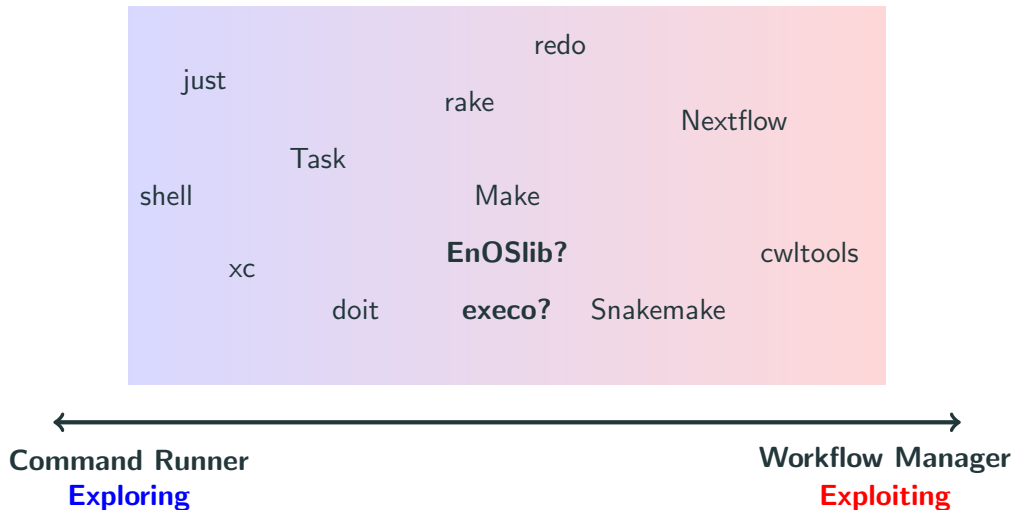
Workflow Managers vs. Command Runners



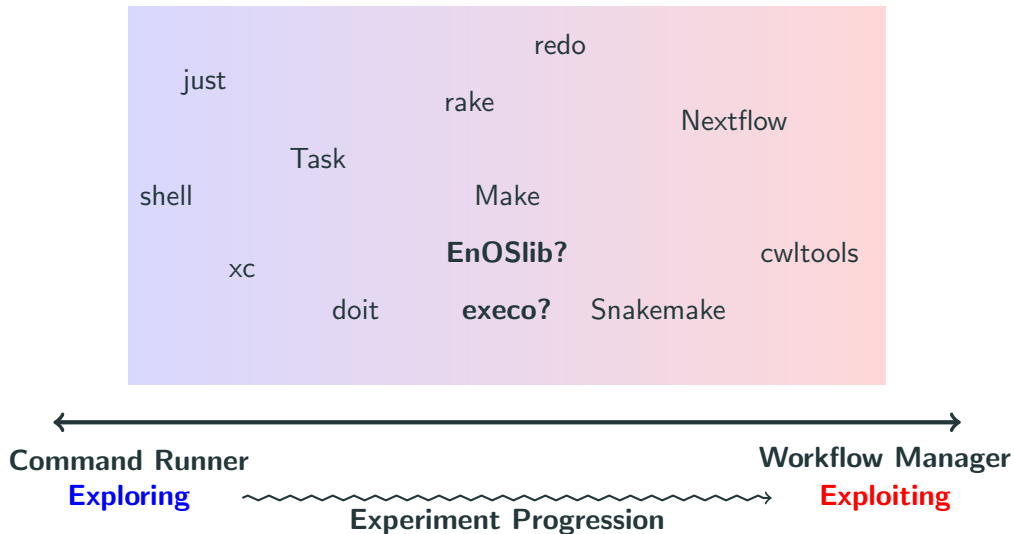
Workflow Managers vs. Command Runners



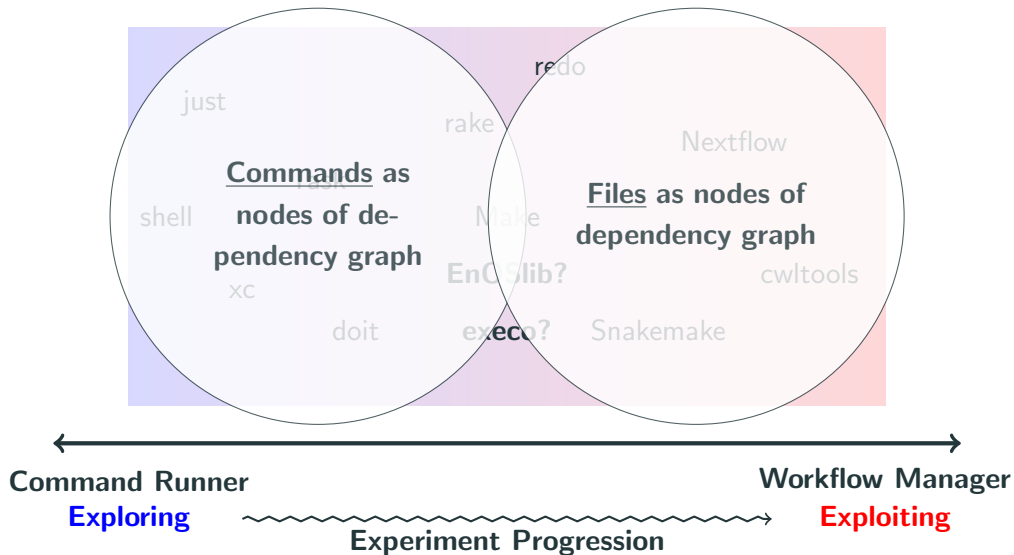
Workflow Managers vs. Command Runners



Workflow Managers vs. Command Runners



Workflow Managers vs. Command Runners



Workflow Managers vs. Command Runners

